



Harmonic Motion

11-1 INTRODUCTION

When a body executes a to-and-fro motion about some fixed point, its motion is said to be *oscillatory*. In this chapter we consider a special type of oscillatory motion called *harmonic motion*. Motion of this sort is closely approximated by that of a body suspended from a spring, by a pendulum swinging with a small amplitude, and by the balance wheel of a watch. The vibrations of the strings and air columns of musical instruments either are harmonic or are a superposition of harmonic motions. Modern atomic theory leads us to believe that the molecules of a solid body oscillate with nearly harmonic motion about their fixed lattice positions, although of course their motion cannot be directly observed.

In every form of wave motion, the particles of the medium in which the wave is traveling oscillate with harmonic motion or with a superposition of such motions. This is true even for light waves and radiowaves in empty space, except that instead of material particles the quantities that oscillate are the electric and magnetic intensities associated with the wave. As a final example, the equations describing the behavior of an electrical circuit in which there is an alternating current have the same form as those for the harmonic motion of a material body. It can be seen that a study of harmonic motion will lay the foundation for future work in many different fields of physics.

11-2 ELASTIC RESTORING FORCES

It has been shown in Chapter 10 that when a body is caused to change its shape, the distorting force is proportional to the amount of the change, provided the proportional limit of elasticity is not exceeded. The

change may be in the nature of an increase in length, as of a rubber band or a coil spring, or a decrease in length or a bending as of a flat spring, or a twisting of a rod about its axis, or of many other forms. The term “force” is to be interpreted liberally as the force, or torque, or pressure, or whatever may be producing the distortion. If we restrict the discussion to the case of a push or a pull, where the distortion is simply the displacement of the point of application of the force, the force and displacement are related by Hooke’s law,

$$F = kx,$$

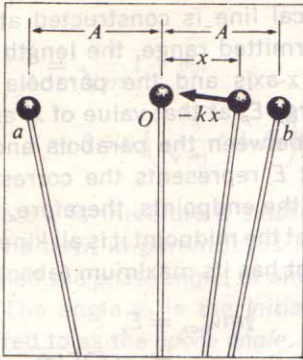
where k is a proportionality constant called the *force constant* and x is the displacement from the equilibrium position.

In this equation, F stands for the force which must be exerted on an elastic body to produce the displacement x . The force with which the elastic body pulls back on an object to which it is attached is called the *restoring force* and is equal to $-kx$.

11-3 DEFINITIONS

To fix our ideas, suppose that a flat strip of steel such as a hacksaw blade is clamped vertically in a vise and a small body is attached to its upper end, as in Fig. 11-1. Assume that the strip is sufficiently long and the displacement sufficiently small so that the motion is essentially along a straight line. The mass of the strip itself is negligible.

Let the top of the strip be pulled to the right a distance A , as in Fig. 11-1, and released. The attached body is then acted on by a restoring force exerted by the steel strip and directed toward the equilibrium position



11-1 Motion under an elastic restoring force.

It therefore accelerates in the direction of this force, and moves in toward the center with increasing speed. The rate of increase (i.e., the acceleration) is not constant, however, since the accelerating force becomes smaller as the body approaches the center.

When the body reaches the center the restoring force has decreased to zero, but because of the velocity which has been acquired, the body "overshoots" the equilibrium position and continues to move toward the left. As soon as the equilibrium position is passed, the restoring force again comes into play, directed now toward the right. The body therefore decelerates, and at a rate which increases with increasing distance from O. It will therefore be brought to rest at some point to the left of O, and repeat its motion in the opposite direction.

Both experiment and theory show that the motion will be confined to a range $\pm A$ on either side of the equilibrium position, each to-and-fro movement taking place in the same length of time. If there were no loss of energy by friction, the motion would continue indefinitely once it had been started. This type of motion, under the influence of an elastic restoring force and in the absence of all friction, is called *simple harmonic motion*, often abbreviated SHM.

Any sort of motion which repeats itself in equal intervals of time is called *periodic*, and if the motion is back and forth over the same path it is also called *oscillatory*.

A *complete vibration* or *complete cycle* means one round trip, say from a to b and back to a, or from O to b and back to O.

The *periodic time*, or simply the *period* of the motion, represented by T , is the time required for one complete vibration.

The *frequency*, f , is the number of complete vibrations per unit time. Evidently the frequency is the reciprocal of the period, or $T = 1/f$. The mks unit, one cycle per second, is called one *hertz* (1 Hz).

The *coordinate*, x , at any instant, is the distance away from the equilibrium position or center of the path at that instant.

The *amplitude*, A , is the maximum coordinate. The total range of the motion is therefore $2A$.

11-4 EQUATIONS OF SIMPLE HARMONIC MOTION

We now wish to find expressions for the coordinate, velocity, and acceleration of a body moving with simple harmonic motion, just as we found those for a body moving with constant acceleration. It must be emphasized that the equations of motion with *constant* acceleration cannot be applied, since the acceleration is continually changing.

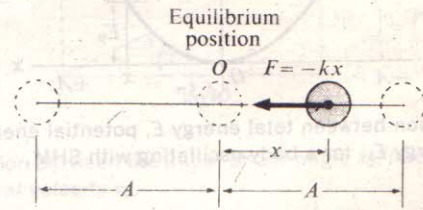


Fig. 11-2.

Figure 11-2 represents the vibrating body of Fig. 11-1 at some instant when its coordinate is x . The resultant force on it is simply the elastic restoring force, $-kx$, and from Newton's second law,

$$F = -kx = ma = m \frac{dv}{dt} \frac{dx}{dt}$$

From the second and fourth terms, we have

$$m \frac{dv}{dx} \frac{dx}{dt} + kx = 0. \quad (11-1)$$

Therefore

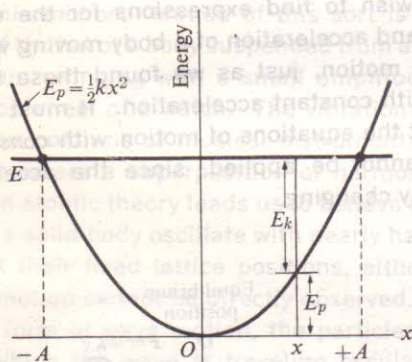
$$\int mv \, dv + \int kx \, dx = 0,$$

which integrates to

$$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = C_1. \quad (11-2)$$

The first term on the left is the kinetic energy E_k of the body and the second is its elastic potential energy E_p . Equation (11-2) therefore states that the total energy of the system is constant, and the integration constant C_1 equals the total energy E (this result is only to be expected, since the system is conservative):

$$E_k + E_p = E.$$



11-3 Relation between total energy E , potential energy E_p , and kinetic energy E_k , for a body oscillating with SHM.

The significance of this relation can be brought out by the graph shown in Fig. 11-3, in which energy is plotted vertically and the coordinate x horizontally. First, the curve representing the potential energy, $E_p = \frac{1}{2}kx^2$, is constructed. (This curve is a parabola.) Next, a horizontal line is drawn at a height equal to the total energy E . We see at once that the motion is restricted to values of x lying between the points at which the horizontal line intersects the parabola, since if x were outside this range the potential energy would exceed the total energy, which is impossible. The motion of the vibrating body is analogous to that of a particle released at a height E on a frictionless track shaped like the potential energy curve, and the motion is said to take place in a "potential energy well."

If a vertical line is constructed at any value within the permitted range, the length of the segment between the x -axis and the parabola represents the potential energy E_p at that value of x , and the length of the segment between the parabola and the horizontal line at height E represents the corresponding kinetic energy E_k . At the endpoints, therefore, the energy is all potential and at the midpoint it is all kinetic. The velocity at the midpoint has its maximum (absolute) value $v_{\max}$.

$$\frac{1}{2}mv_{\max}^2 = E,$$

$$v_{\max} = \pm\sqrt{2E/m}, \quad (11-3)$$

the sign of v being positive or negative depending on the direction of motion.

At the ends of the path the coordinate has its maximum (absolute) value $x_{\max}$, equal to the amplitude A :

$$\frac{1}{2}kx_{\max}^2 = E, \quad x_{\max} = \pm\sqrt{2E/k},$$

$$A = |x_{\max}| = \sqrt{2E/k}. \quad (11-4)$$

The velocity v at any coordinate x , from Eq. (11-2)

$$v = \pm\sqrt{\frac{2E - kx^2}{m}}. \quad (11-5)$$

Making use of Eq. (11-4), this can be written

$$v = \sqrt{k/m}\sqrt{A^2 - x^2}. \quad (11-6)$$

The expression for the coordinate x as a function of time can now be obtained by replacing v with dx/dt in Eq. (11-6), and integrating. This gives

$$\begin{aligned} \int \frac{dx}{\sqrt{A^2 - x^2}} &= \sqrt{\frac{k}{m}} \int dt, \\ \sin^{-1} \frac{x}{A} &= \sqrt{\frac{k}{m}} t + C_2. \end{aligned} \quad (11-7)$$

Let x_0 be the value of x when $t = 0$. The integration constant C_2 is then

$$C_2 = \sin^{-1} \frac{x_0}{A}.$$

That is, C_2 is the angle (in radians) whose sine equals x_0/A . Let us represent this angle by θ_0 :

$$\sin \theta_0 = \frac{x_0}{A}, \quad \theta_0 = \sin^{-1} \frac{x_0}{A}. \quad (11-8)$$

Equation (11-7) can now be written

$$\sin^{-1} \frac{x}{A} = \sqrt{\frac{k}{m}} t + \theta_0, \quad (11-9)$$

$$x = A \sin \left(\sqrt{\frac{k}{m}} t + \theta_0 \right).$$

The coordinate x is therefore a *sinusoidal* function of the time t . The term in parentheses is an *angle*, in radians. It is called the *phase angle*, or simply the *phase* of the motion. The angle θ_0 is the *initial phase angle*, and is also referred to as the *epoch angle*.

The period T is the time required for one complete oscillation. That is, the coordinate x has the same value at times t and $t + T$. In other words, the phase angle $\left(\sqrt{\frac{k}{m}} t + \theta_0 \right)$ increases by 2π radians in the time T :

$$\sqrt{\frac{k}{m}} (t + T) + \theta_0 = \left(\sqrt{\frac{k}{m}} t + \theta_0 \right) + 2\pi,$$

$$T = 2\pi \sqrt{\frac{m}{k}}. \quad (11-10)$$

The period T therefore depends only on the mass m and the force constant k . It does not depend on the amplitude (or on the total energy). For given values of m and k , the time of one complete oscillation is the same whether the amplitude is large or small. A motion that has this property is said to be *isochronous* (equal time). The frequency f , or the number of complete oscillations per unit time, is the reciprocal of the period T :

$$f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}.$$

The *angular frequency* ω is defined as $\omega = 2\pi f$ and is expressed in radians per second. It follows from the preceding equations that

$$\omega = \sqrt{k/m},$$

Eqs. (11-9) and (11-6) can be written more compactly as

$$x = A \sin (\omega t + \theta_0), \quad (11-11)$$

$$v = \omega \sqrt{A^2 - x^2}. \quad (11-12)$$

Expressions for the velocity and acceleration as functions of time can now be obtained by differentiation:

$$v = \frac{dx}{dt} = \omega A \cos (\omega t + \theta_0), \quad (11-13)$$

$$a = \frac{dv}{dt} = -\omega^2 A \sin (\omega t + \theta_0). \quad (11-14)$$

Since $A \sin (\omega t + \theta_0) = x$, the expression for the acceleration as a function of x is

$$a = -\omega^2 x. \quad (11-15)$$

It follows from Eq. (11-13) that if v_0 is the velocity when $t = 0$, then

$$\cos \theta_0 = \frac{v_0}{\omega A}. \quad (11-16)$$

This equation, together with Eq. (11-8), $\sin \theta_0 = x_0/A$, completely determines the initial phase angle θ_0 . That is, the angle θ_0 depends both on the initial position x_0 and the initial velocity v_0 .

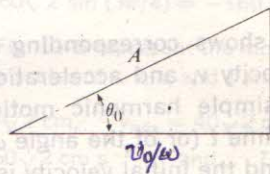
11-4 Relation between the initial phase angle θ_0 , initial position x_0 , and initial velocity v_0 .

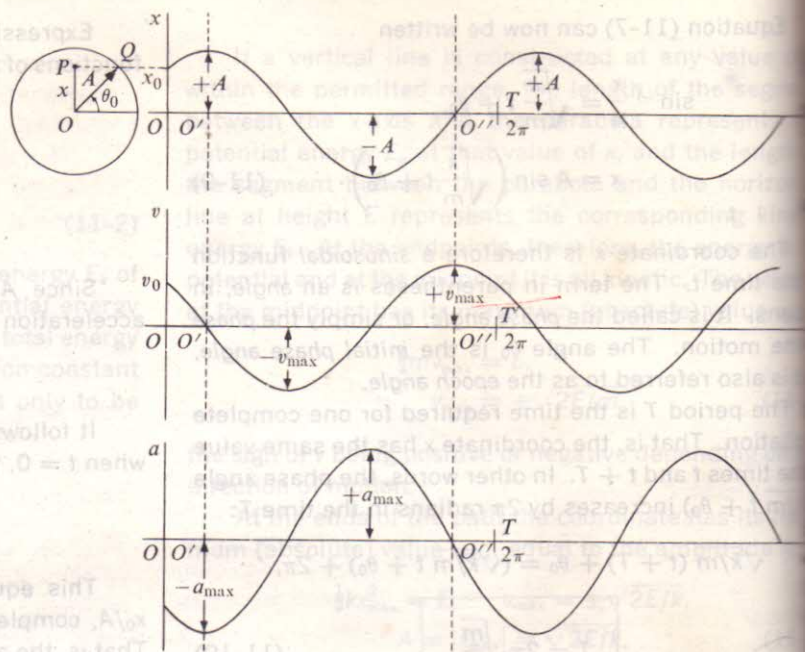
Figure 11-4 shows how the initial phase angle depends on the initial position and initial velocity. It follows from this triangle that

$$A = \sqrt{x_0^2 + (v_0/\omega)^2}, \quad (11-17)$$

a relation that can also be obtained by setting the initial total energy, $\frac{1}{2} k x_0^2 + \frac{1}{2} m v_0^2$, equal to the potential energy at maximum displacement. The motion is therefore completely determined, for given values of m and k , when the initial position and initial velocity are known.

The equations of simple harmonic motion may be summarized by comparing them with similar equations





11-5 Graphs of coordinate x , velocity v , and acceleration a of a body oscillating with simple harmonic motion.

for rectilinear motion with constant acceleration (Table 11-1).

Figure 11-5 shows corresponding graphs of the coordinate x , velocity v , and acceleration a of a body oscillating with simple harmonic motion, plotted as functions of the time t (or of the angle ωt). The initial coordinate is x_0 and the initial velocity is v_0 . The angle

θ_0 has been taken as $\pi/4$ rad. Each curve repeats in a time interval equal to the period T , during which angle ωt increases by 2π rad. Note that when the body is at either end of its path, i.e., when the coordinate has its maximum positive or negative value ($\pm A$), the velocity is zero and the acceleration has its maximum negative or positive value ($\mp a_{\max}$). Note also that when the body passes through its equilibrium position ($x = 0$), the velocity has its maximum positive or negative value ($\pm v_{\max}$) and the acceleration is zero.

The equations of motion take a simpler form if we set $t = 0$ when the body is at the midpoint, or is at either end of its path. For example, suppose we let $t = 0$ when the body has its maximum positive displacement. Then $x_0 = +A$, $\sin \theta_0 = 1$, $\theta_0 = \pi/2$, and

$$\begin{aligned} x &= A \sin(\omega t + \pi/2) = A \cos \omega t, \\ v &= -\omega A \sin \omega t, \\ a &= -\omega^2 A \cos \omega t. \end{aligned} \quad (11-1)$$

This corresponds to moving the origin from point O , in Fig. 11-5, to the point O' . The graph of x versus t becomes a cosine curve, that of v versus t becomes a negative sine curve, and that of a a negative cosine curve.

TABLE 11-1

Rectilinear motion with constant acceleration	Simple harmonic motion (in terms of ω and θ_0)
$a = \text{constant}$	$a = -\omega^2 x$ $a = -\omega^2 A \sin(\omega t + \theta_0)$
$v^2 = v_0^2 + 2a(x - x_0)$ $v = v_0 + at$	$v = \pm \omega \sqrt{A^2 - x^2}$ $v = \omega A \cos(\omega t + \theta_0)$
$x = x_0 + v_0 t + \frac{1}{2}at^2$	$x = A \sin(\omega t + \theta_0)$
$\omega = 2\pi/T = 2\pi f = \sqrt{k/m}$, $\sin \theta_0 = x_0/A$, $\cos \theta_0 = v_0/\omega A$.	$A = \sqrt{x_0^2 + (v_0/\omega)^2}$

If we set $t = 0$ when the body is at the midpoint and moving toward the right, then

$$x_0 = 0, \quad \sin \theta_0 = 0, \quad \theta_0 = 0$$

$$\begin{aligned} x &= A \sin \omega t, \\ v &= \omega A \cos \omega t, \\ a &= -\omega^2 A \sin \omega t. \end{aligned} \quad (11-19)$$

This corresponds to displacing the origin to the point O'' in Fig. 11-5.

Let the mass of the body in Fig. 11-2 be 25 g, the force constant k be 400 dyn cm^{-1} , and let the motion be started by displacing the body 10 cm to the right of its equilibrium position and imparting to it a velocity toward the right of 40 cm s^{-1} . Compute (a) the period T , (b) the frequency f , (c) the angular frequency ω , (d) the total energy E , (e) the amplitude A , (f) the phase angle θ_0 , (g) the maximum velocity $v_{\max}$, (h) the maximum acceleration $a_{\max}$, (i) the coordinate, velocity, and acceleration at time $\pi/8$ s after the start of the motion.

$$T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{25 \text{ g}}{400 \text{ dyn cm}^{-1}}} = \frac{\pi}{2} \text{ s} = 1.57 \text{ s}.$$

$$f = \frac{1}{T} = \frac{2}{\pi} \text{ Hz} = 0.638 \text{ Hz}.$$

$$\omega = 2\pi f = 4 \text{ rad s}^{-1}.$$

$$E = \frac{1}{2}mv_0^2 + \frac{1}{2}kx_0^2 = 40,000 \text{ ergs}.$$

$$A = \sqrt{2E/k} = 10\sqrt{2} \text{ cm}.$$

$$\theta_0 = x_0/A = 1/\sqrt{2}, \quad \theta_0 = \pi/4 \text{ rad}.$$

$$v_{\max} = \sqrt{2E/m} = 40\sqrt{2} \text{ cm s}^{-1} = 56.6 \text{ cm s}^{-1}.$$

The maximum velocity occurs at the midpoint, where $x = 0$.

Hence, from Eq. (11-12),

$$|v_{\max}| = \omega A = 40\sqrt{2} \text{ cm s}^{-1}.$$

h) The maximum acceleration occurs at the ends of the path where the force is a maximum. From Eq. (11-15),

$$|a_{\max}| = \omega^2 x_{\max} = 160\sqrt{2} \text{ cm s}^{-2}.$$

i) The equations of motion are:

$$x = 10\sqrt{2} \sin\left(4t + \frac{\pi}{4}\right),$$

$$v = 40\sqrt{2} \cos\left(4t + \frac{\pi}{4}\right),$$

$$a = -160\sqrt{2} \sin\left(4t + \frac{\pi}{4}\right).$$

When $t = \pi/8$ s, the phase angle is

$$\left(4t + \frac{\pi}{4}\right) = \frac{3\pi}{4} \text{ rad},$$

$$x = 10\sqrt{2} \sin(3\pi/4) = 10 \text{ cm},$$

$$v = 40\sqrt{2} \cos(3\pi/4) = -40 \text{ cm s}^{-1},$$

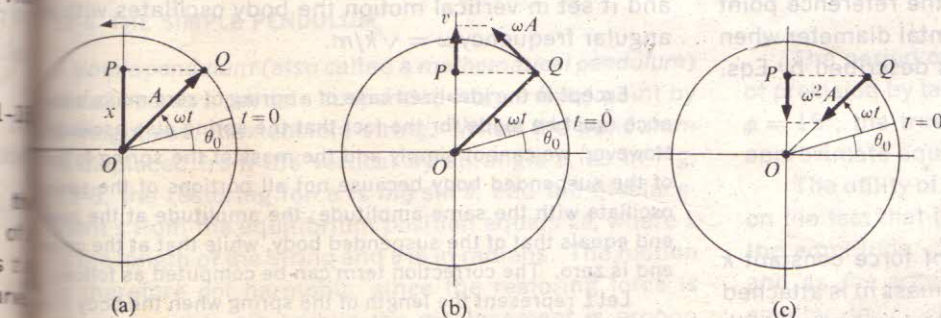
$$a = -160\sqrt{2} \sin(3\pi/4) = -160 \text{ cm s}^{-2}.$$

The curves of Fig. 11-5 represent the motion of the body in this example, if the scales of x , v , a , and t are such that

$$A = 10\sqrt{2} \text{ cm}, \quad v_{\max} = 40\sqrt{2} \text{ cm s}^{-1},$$

$$a_{\max} = 160\sqrt{2} \text{ cm s}^{-2}, \quad \text{and} \quad T = \pi/2 \text{ s}.$$

The equations of simple harmonic motion can be given a geometrical interpretation as follows. Let the line segment OQ in Fig. 11-6(a), of length equal to the amplitude A , rotate with an angular velocity ω about the fixed point O . The rotating line segment is often referred



11-6 Representation of simple harmonic motion by the projection onto the vertical axis of the tip of rotor OQ .

to as a *rotating vector*, but strictly speaking it is not a vector quantity. That is, while it has a specified direction in a diagram, it has no specified direction in space. It is better described as a *rotor*. (The German term is "Zeiger," meaning a pointer in the sense of a clock hand or the pointer on a pressure gauge.) Let the rotor OQ make an angle with the horizontal axis, at time $t = 0$, equal to the initial phase angle θ_0 . Point P is the projection of point Q onto the vertical axis, and as OQ rotates, point P oscillates along this axis.

We now show that the equations of motion of P are the same as those of a body oscillating with simple harmonic motion of amplitude A , angular frequency ω , and initial phase angle θ_0 . Let x represent the length of OP . At any time t , the angle between the radius OQ and the horizontal axis equals the phase angle $\omega t + \theta_0$ and

$$x = A \sin(\omega t + \theta_0).$$

The velocity of point Q (see Fig. 11-6b) is ωA , and its vertical component, equal to the velocity of P , is

$$v = \omega A \cos(\omega t + \theta_0).$$

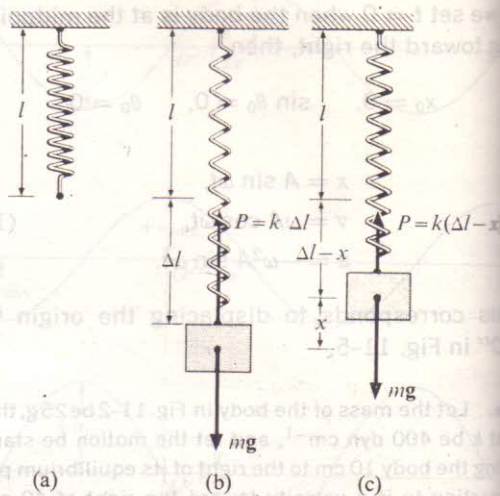
The acceleration of Q is its radial acceleration $\omega^2 A$ (see Fig. 11-6c) and its vertical component, equal to the acceleration of P , is

$$a = -\omega^2 A \sin(\omega t + \theta_0).$$

The negative sign must be included because the acceleration is negative whenever the sine of the phase angle is positive, and vice versa. These equations are, of course, just the general equations of harmonic motion. In the special case corresponding to Eqs. (11-18), the initial phase angle is 90° and the reference point Q is at the top of the circle when $t = 0$. If the reference point is at the right-hand end of the horizontal diameter when $t = 0$, then $\theta_0 = 0$ and the motion is described by Eqs. (11-19).

11-5 MOTION OF A BODY SUSPENDED FROM A COIL SPRING

Figure 11-7(a) shows a coil spring of force constant k and no-load length ℓ . When a body of mass m is attached to the spring as in part (b), it hangs in equilibrium with the spring extended by an amount $\Delta\ell$ such that the



11-7 The restoring force on a body suspended by a spring is proportional to the coordinate measured from the equilibrium position.

upward force P exerted by the spring is equal to weight of the body, mg . But $P = k\Delta\ell$, so

$$k\Delta\ell = mg.$$

Now suppose the body is at a distance x above equilibrium position, as in part (c). The extension of the spring is now $\Delta\ell - x$, the upward force it exerts on the body is $k(\Delta\ell - x)$, and the resultant force F on the body is

$$F = k(\Delta\ell - x) - mg = -kx.$$

The resultant force is therefore proportional to the displacement of the body from its equilibrium position and if set in vertical motion the body oscillates with angular frequency $\omega = \sqrt{k/m}$.

Except in the idealized case of a spring of zero mass, allowance must be made for the fact that the spring also oscillates. However, we cannot simply add the mass of the spring to that of the suspended body because not all portions of the spring oscillate with the same amplitude; the amplitude at the lower end equals that of the suspended body, while that at the upper end is zero. The correction term can be computed as follows.

Let L represent the length of the spring when the body is at its equilibrium position, and let m_s be the mass of the spring. Let us calculate the kinetic energy of the spring at an instant

the velocity of the lower end is v . Consider an element of the spring of length dy , at a distance y below the fixed upper end. The mass of the element, dm_s , is

$$dm_s = \frac{m_s}{L} dy.$$

portions of the spring will be assumed to oscillate in phase and the velocity of the element, v_s , to be proportional to its distance from the fixed end: $v_s = (y/L)v$.

The kinetic energy of the element is

$$dE_k = \frac{1}{2} \cdot dm_s \cdot v_s^2 = \frac{1}{2} \cdot \frac{m_s}{L} dy \cdot \left(\frac{y}{L}v\right)^2,$$

and the total kinetic energy of the spring is

$$E_k = \frac{1}{2} \cdot \frac{m_s v^2}{L^3} \cdot \int_0^L y^2 dy = \frac{1}{2} \left(\frac{1}{3} m_s\right) v^2.$$

This equals the kinetic energy of a body of mass one-third the mass of the spring, moving with the same velocity as that of the suspended body. In other words, the equivalent mass of the vibrating system equals that of the suspended body plus one-third the mass of the spring.

Example. A body of mass 1 kg is suspended from a coil spring whose mass is 0.09 kg and whose force constant is 66 N m^{-1} . Find the frequency and amplitude of the ensuing motion if the body is displaced 0.03 m below its equilibrium position and given a downward velocity of 0.4 m s^{-1} . The angular frequency

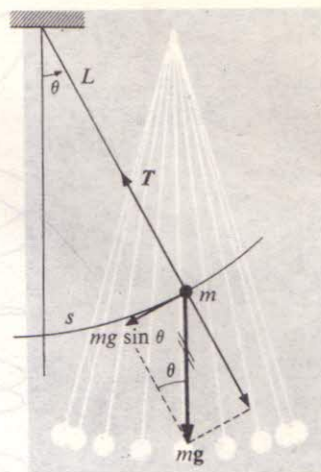
$$\omega = \sqrt{\frac{k}{m + m_s/3}} = \sqrt{\frac{66 \text{ N m}^{-1}}{1.03 \text{ kg}}} = 8.00 \text{ rad s}^{-1}.$$

The amplitude is expressed in terms of the initial coordinate and velocity by means of Eq. (11-17). Thus,

$$A = \sqrt{x_0^2 + (v_0/\omega)^2} = \sqrt{(0.03 \text{ m})^2 + (0.4/8)^2} = 0.0582 \text{ m}.$$

11-6 THE SIMPLE PENDULUM

A **simple pendulum** (also called a **mathematical pendulum**) is defined as a particle suspended from a fixed point by a weightless, inextensible string. When the pendulum is displaced from the vertical by an angle θ , as in Fig. 11-8, the restoring force is $mg \sin \theta$, and the displacement s from the equilibrium position equals $L\theta$, where L is the length of the string and θ is in radians. The motion is therefore *not* harmonic, since the restoring force is proportional to $\sin \theta$ while the displacement is propor-



11-8 Forces on the bob of a simple pendulum.

tional to θ . However, if the angle θ is small we can approximate $\sin \theta$ by θ , and the restoring force is then

$$F \approx -mg\theta \approx -\left(\frac{mg}{L}\right)s.$$

The effective force constant of the pendulum is therefore $k = mg/L$, and the period is

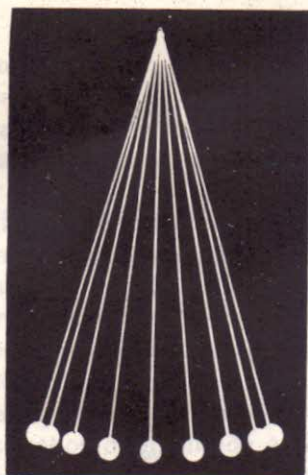
$$T \approx 2\pi\sqrt{m/k} \approx 2\pi\sqrt{L/g}. \quad (11-20)$$

It can be shown that the exact equation for the period, when the maximum angular displacement is ϕ , is given by the infinite series

$$T = 2\pi\sqrt{\frac{L}{g}} \left(1 + \frac{1^2}{2^2} \sin^2 \frac{\phi}{2} + \frac{1^2 \cdot 3^2}{2^2 \cdot 4^2} \sin^4 \frac{\phi}{2} + \dots\right). \quad (11-21)$$

The period can be computed to any desired degree of precision by taking enough terms in the series. When $\phi = 15^\circ$, the true period differs from that given by the approximate equation (11-20) by less than 0.5%.

The utility of the pendulum as a timekeeper is based on the fact that the period is practically independent of the amplitude. Thus, as a pendulum clock runs down and as the amplitude of the swings becomes slightly smaller, the clock will still keep very nearly correct time.



11-9 A single swing of a simple pendulum.

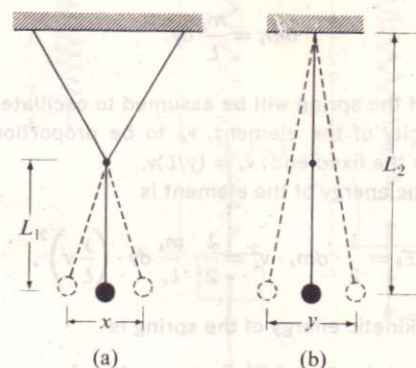
The simple pendulum is also a precise and convenient method of measuring the acceleration of gravity, g , without actually resorting to free fall, since L and T may readily be measured. More complicated pendulums find considerable application in the field of geophysics. Local deposits of ore or oil, if their density differs from that of their surroundings, affect the local value of g , and precise measurements of this quantity over an area which is being prospected often furnish valuable information regarding the nature of underlying deposits.

Figure 11-9 is a multiframe photograph of a single swing of a simple pendulum.

11-7 LISSAJOUS FIGURES

The curves known as *Lissajous figures* are the paths traced out by a particle which is oscillating simultaneously in two mutually perpendicular directions. In general, the amplitude and frequency may be different in the two directions, and the two oscillations may have an arbitrary initial phase difference.

The pendulum bob in Fig. 11-10, supported by three cords forming a Y, illustrates one means of producing this type of motion. When vibrating in the x -direction, as in (a), the frequency is that of a simple pendulum of length L_1 . In the y -direction the frequency is that of a pendulum of length L_2 . If displaced in both the x - and



11-10 Double pendulum for producing Lissajous figures.

y -directions, and released, the bob executes vibrations of both frequencies simultaneously.

The spot on the screen of a cathode-ray tube, produced by the impact of a rapidly moving stream of electrons, will also move in a Lissajous figure if sinusoidal alternating voltages are applied simultaneously to the horizontal and vertical deflecting plates.

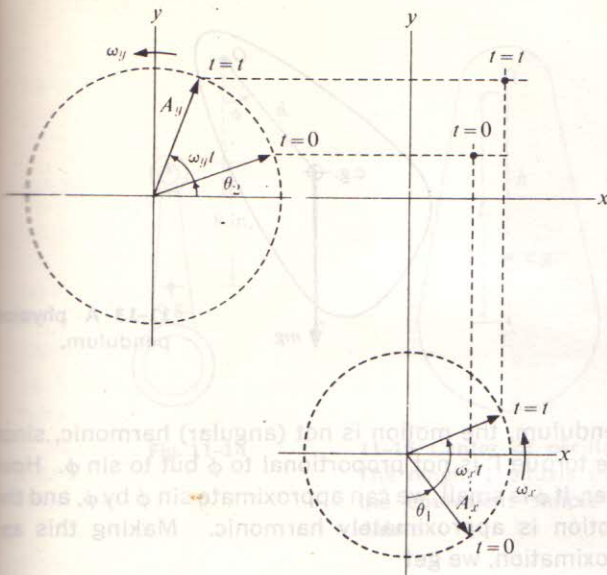
The general expressions for the x - and y -coordinates of the oscillating particle are

$$x = A_x \sin(\omega_x t + \theta_1), \quad y = A_y \sin(\omega_y t + \theta_2),$$

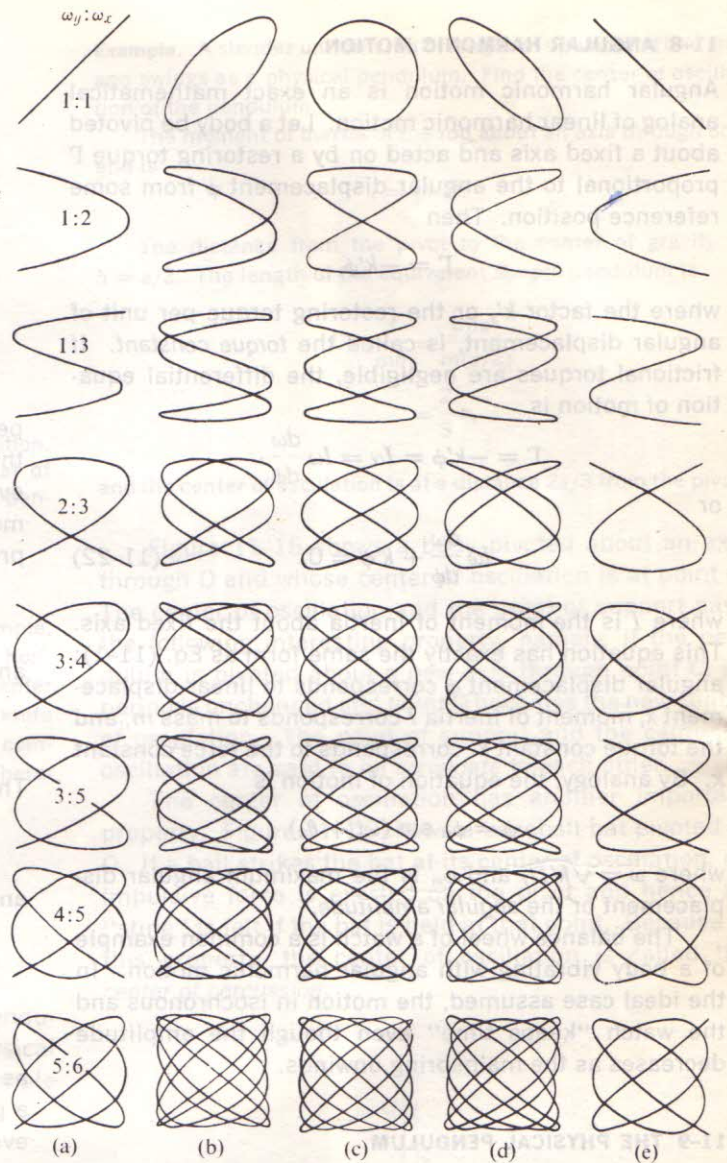
where A_x and A_y are the respective amplitudes, ω_x and ω_y the corresponding angular frequencies, and θ_1 and θ_2 the initial phase angles. These are the equations of the path in parametric form.

The equations above are represented graphically by the rotor diagrams of Fig. 11-11. The x -coordinate of the tip of the rotor in the lower diagram gives the x -coordinate of the oscillating particle, and the y -coordinate of the tip of the rotor in the upper diagram gives its y -coordinate. Hence by projecting up, and across, from the tips of these rotors, the position of the particle at any time can be determined. The diagram shows its position at time $t = 0$ and at some arbitrary later time t .

The curves in Fig. 11-12 are a few Lissajous figures for various frequency ratios and initial phase differences ($\theta_2 - \theta_1$). The amplitudes A_x and A_y are equal in each



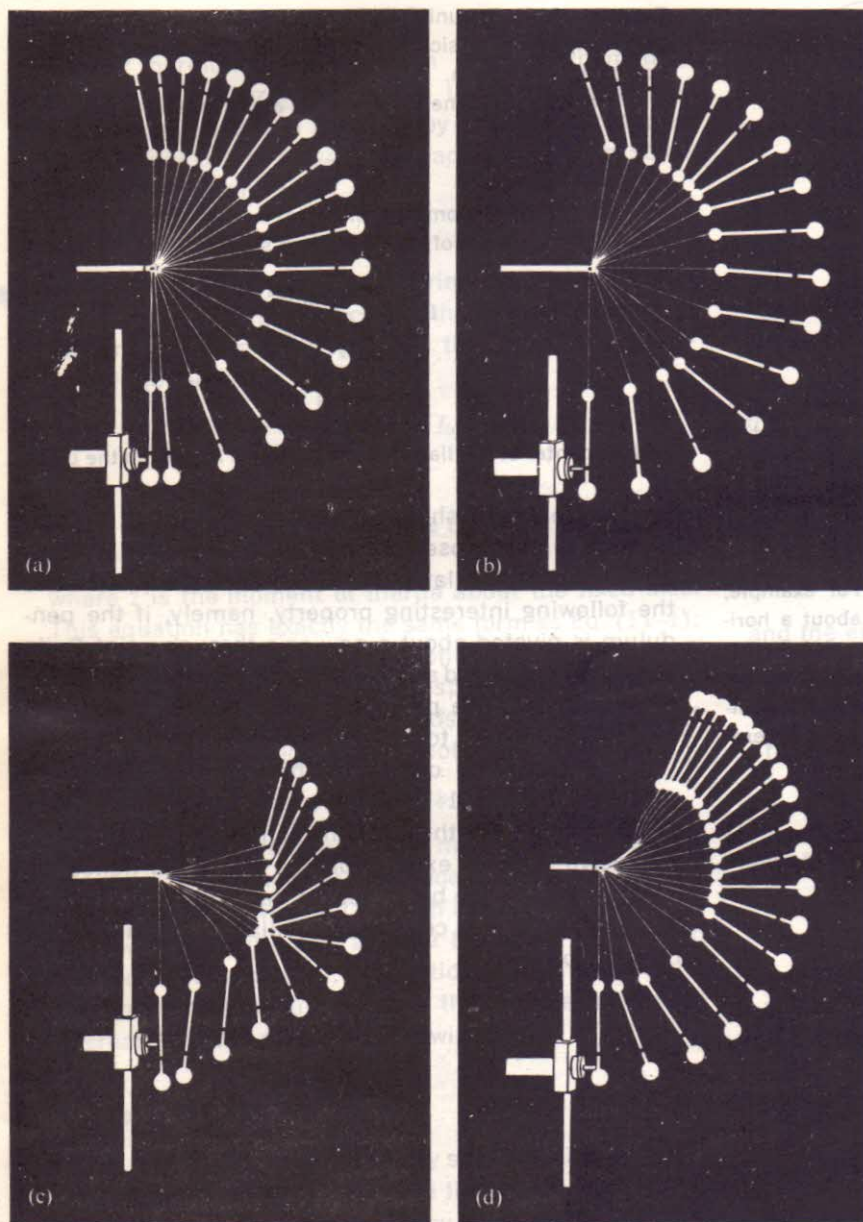
11-11 Graphical construction for Lissajous figures.



11-12 Lissajous figures.

case. If the frequencies are commensurable, as in the figures shown, the particle retraces a closed path over and over. If they are not, the path does not close on itself, and the pattern may be extremely complicated. If the frequencies are very nearly commensurable, the path changes very slowly, and if the motion itself is rapid, as is often the case when the figures are formed on an oscilloscope,

the impression is that of a closed curve which gradually alters its form. Thus, if the frequencies are very nearly equal ($\omega_x/\omega_y \approx 1$), the path changes slowly from a straight line at 45° , as in Fig. 11-12(a), to an ellipse as in (b), then to a circle as in (c), then to an ellipse as in (d), with its major axis at right angles to that in (b), then to a straight line as in (e), etc.



In Fig. 11-17, a series of multiflash photographs, the center of gravity is marked by a black band. In (a), the body is struck at its center of percussion relative to a pivot at the upper end of the cord, and it starts to swing smoothly about this pivot. In (b), the body is struck at

its center of gravity. Note that it does not start to rotate about the pivot, but that its initial motion is one of pure translation. That is, the center of percussion does not coincide with the center of gravity. In (c), the body is struck above, and in (d) below its center of percussion.

11-17 The motion of a body suspended by a cord when the body is struck a horizontal blow.

Problems

11-1 The general equation of simple harmonic motion,

$$y = A \sin(\omega t + \theta_0),$$

can be written in the equivalent form

$$y = B \sin \omega t + C \cos \omega t.$$

(a) Find the expressions for the amplitudes B and C in terms of the amplitude A and the initial phase angle θ_0 . (b) Interpret these expressions in terms of a rotating vector diagram.

11-2 A body of mass 0.25 kg is acted on by an elastic restoring force of force constant $k = 25 \text{ N m}^{-1}$. (a) Construct the graph of elastic potential energy E_p as a function of displacement x , over a range of x from -0.3 m to $+0.3 \text{ m}$. Let $1 \text{ in.} = 0.25 \text{ J}$ vertically, and $1 \text{ in.} = 0.1 \text{ m}$ horizontally.

The body is set into oscillation with an initial potential energy of 0.6 J and an initial kinetic energy of 0.2 J . Answer the following questions by reference to the graph: (b) What is the amplitude of oscillation? (c) What is the potential energy when the displacement is one-half the amplitude? (d) At what displacement are the kinetic and potential energies equal? (e) What is the speed of the body at the midpoint of its path? Find (f) the period T , (g) the frequency f , and (h) the angular frequency ω . (i) What is the initial phase angle θ_0 if the amplitude $A = 15 \text{ cm}$, the initial displacement $x_0 = 7.5 \text{ cm}$, and the initial velocity v_0 is negative?

11-3 A body is vibrating with simple harmonic motion of amplitude 15 cm and frequency 4 Hz . Compute (a) the maximum values of the acceleration and velocity, (b) the acceleration and velocity when the coordinate is 9 cm , (c) the time required to move from the equilibrium position to a point 12 cm distant from it.

11-4 A body of mass 10 g moves with simple harmonic motion of amplitude 24 cm and period 4 s . The coordinate is $+24 \text{ cm}$ when $t = 0$. Compute (a) the position of the body when $t = 0.5 \text{ s}$, (b) the magnitude and direction of the force acting on the body when $t = 0.5 \text{ s}$, (c) the minimum time required for the body to move from its initial position to the point where $x = -12 \text{ cm}$, (d) the velocity of the body when $x = -12 \text{ cm}$.

11-5 The motion of the piston of an automobile engine is approximately simple harmonic. (a) If the stroke of an engine (twice the amplitude) is 4 in. and the angular velocity is $3600 \text{ rev min}^{-1}$, compute the acceleration of the piston at the end of its stroke. (b) If the piston weighs 1 lb , what resultant force must be exerted on it at this point? (c) What is the velocity of the piston, in miles per hour, at the midpoint of its stroke?

11-6 A body whose weight is 4 lb is suspended from a spring of negligible mass, and is found to stretch the spring 8 in. (a) What is the force constant of the spring? (b) What is the period of oscillation of the body, if pulled down and released? (c) What would be the period of a body weighing 8 lb , hanging from the same spring?

11-7 The scale of a spring balance reading from zero to 32 lb is 6 in. long. A body suspended from the balance is observed to oscillate vertically at 1.5 Hz . What is the weight of the body? Neglect the mass of the spring.

11-8 A body weighing 8 lb is attached to a coil spring and oscillates vertically in simple harmonic motion. The amplitude is 2 ft , and at the highest point of the motion the spring has its natural unstretched length. Calculate the elastic potential energy of the spring, the kinetic energy of the body, its gravitational potential energy relative to the lowest point of the motion, and the sum of these three energies, when the body is (a) at its lowest point, (b) at its equilibrium position, and (c) at its highest point.

11-9 A load of 320 lb suspended from a wire whose unstretched length l_0 is 10 ft is found to stretch the wire by 0.12 in. The cross-sectional area of the wire, which can be assumed constant, is 0.016 in^2 . (a) If the load is pulled down a small additional distance and released, find the frequency at which it will vibrate. (b) Compute Young's modulus for the wire.

11-10 A small block is executing simple harmonic motion in a horizontal plane with an amplitude of 10 cm . At a point 6 cm away from equilibrium the velocity is $\pm 24 \text{ cm s}^{-1}$. (a) What is the period? (b) What is the displacement when the velocity is $\pm 12 \text{ cm s}^{-1}$? (c) If a small object placed on the oscillating block is just on the verge of slipping at the endpoint of the path, what is the coefficient of friction?

11-11 A force of 6 lb stretches a vertical spring 9 in. (a) What weight must be suspended from the spring so that the system will oscillate with a period of $\pi/4 \text{ s}$? (b) If the amplitude of the motion is 3 in. , where is the body and in what direction is it moving $\pi/12 \text{ s}$ after it has passed the equilibrium position, moving downward? (c) What force does the spring exert on the body when it is 1.8 in. below the equilibrium position, moving upward?

11-12 A body of mass m is suspended from a coil spring and the time for 100 complete oscillations is measured for the following values of m :

$m \text{ (g)}$	100	200	400	1000
Time of 100 oscillations (s)	23.4	30.6	41.8	64.7

Plot graphs of the measured values of (a) T vs. m , and (b) T^2 vs. m . (c) Are the experimental results in agreement with theory? (d) Is either graph a straight line? (e) Does the straight line pass through the origin? (f) What is the force constant of the spring? (g) What is the mass of the spring?

11-13 A body of mass 100 g hangs from a long spiral spring. When pulled down 10 cm below its equilibrium position and released, it vibrates with a period of 2 s. (a) What is its velocity as it passes through the equilibrium position? (b) What is its acceleration when it is 5 cm above the equilibrium position? (c) When it is moving upward, how long a time is required for it to move from a point 5 cm below its equilibrium position to a point 5 cm above it? (d) How much will the spring shorten if the body is removed?

11-14 A body whose mass is 4.9 kg hangs from a spring and oscillates with a period of 0.5 s. How much will the spring shorten when the body is removed?

11-15 Four passengers whose combined weight is 600 lb are observed to compress the springs of an automobile by 2 in. when they enter the automobile. If the total load supported by the springs is 1800 lb, find the period of vibration of the loaded automobile.

11-16 (a) A block suspended from a spring vibrates with simple harmonic motion. At an instant when the displacement of the block is equal to one-half the amplitude, what fraction of the total energy of the system is kinetic and what fraction is potential? (b) When the block is in equilibrium, the length of the spring is an amount s greater than in the unstretched state. Prove that $T = 2\pi\sqrt{s/g}$.

11-17 (a) With what additional force must a vertical spring carrying an 8-lb body in equilibrium be stretched so that, when released, it will perform 48 complete oscillations in 32 s with an amplitude of 3 in.? (b) What force is exerted by the spring on the body when it is at the lowest point, the middle, and the highest point of the path? (c) What is the kinetic energy of the system when the body is 1 in. below the middle of the path? its potential energy?

11-18 A force of 12 lb stretches a certain spring 9 in. A body weighing 8 lb is hung from this spring and allowed to come to rest. It is then pulled down 4 in. and released. (a) What is the period of the motion? (b) What is the magnitude and direction of the acceleration of the body when it is 2 in. above the equilibrium position, moving upward? (c) What is the tension in the spring when the body is 2 in. above the equilibrium position? (d) What is the shortest time required to go from the equilibrium position to the point 2 in. above? (e) If a small object were placed on the oscillating body would it remain or leave? (f) If a small object were placed on the oscillating body and its

amplitude doubled, where would the object and the oscillating body begin to separate?

11-19 Two springs with different force constants k_1 and k_2 are attached to a block of mass m on a level frictionless surface. Calculate the effective force constant in each of the three cases (a), (b), and (c) depicted in Fig. 11-18. (d) A body of mass m suspended from a spring with a force constant k , vibrates with frequency f_1 . When the spring is cut in half and the same body is suspended from one of the halves, the frequency is f_2 . What is the ratio of f_2/f_1 ?

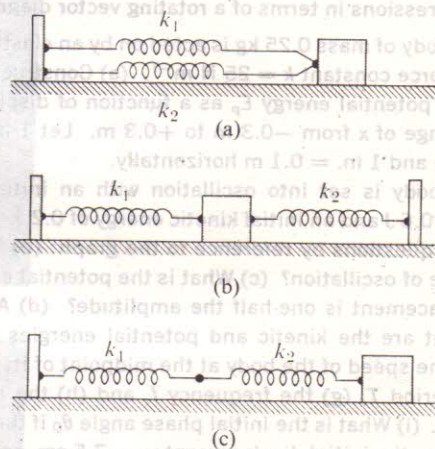


Fig. 11-18

11-20 Two springs, each of unstretched length 20 cm but having different force constants k_1 and k_2 , are attached to opposite ends of a block of mass m on a level frictionless surface. The outer ends of the springs are now attached to two pins P_1 and P_2 , 10 cm from the original positions of the springs. Let $k_1 = 1000 \text{ dyn cm}^{-1}$, $k_2 = 3000 \text{ dyn cm}^{-1}$, $m = 100 \text{ g}$. (See Fig. 11-19.) (a) Find the length of each spring when the block is in its new equilibrium position, after the springs have been attached to the pins. (b) Find the period of vibration of the block if it is slightly displaced from its new equilibrium position and released.

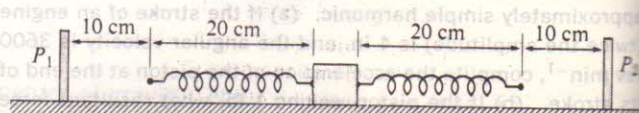


Fig. 11-19

11-21 The block in Problem 11-20 and Fig. 11-19 is oscillating with an amplitude of 5 cm. At the instant it passes through its equilibrium position, a lump of putty of mass 100 g is dropped vertically onto the block and sticks to it. (a) Find the new period and amplitude. (b) Was there a loss of energy and, if so, where did it go? (c) Would the answers be the same if the putty had been dropped on the block when it was at one end of its path?

11-22 A simple pendulum 8 ft long swings with an amplitude of 1 ft. (a) Compute the velocity of the pendulum at its lowest point. (b) Compute its acceleration at the ends of its path.

11-23 Find the length of a simple pendulum whose period is exactly 1 s at a point where $g = 32.2 \text{ ft s}^{-2}$.

11-24 (a) What is the change dT in the period of a simple pendulum when the acceleration due to gravity changes by dg ? (b) What is the fractional change in period, dT/T , in terms of the fractional change dg/g ? (c) A pendulum clock which keeps correct time at a point where $g = 980.0 \text{ cm sec}^{-2}$ is found to lose 10 s day^{-1} at a higher altitude. Use the result of part (b) above to find the value of g at the new location, approximating the differentials dT and dg by the small finite changes in T and g .

(This problem illustrates a useful technique when it is desired to find the new value of a quantity, given a small change in another quantity on which the first depends. It enables one to compute the *change* in the quantity instead of computing the new value directly. To compute the new value of g directly to within 0.1 cm s^{-2} , computations must be carried out to four significant figures, which is beyond the precision of an ordinary slide rule and requires the use of four-place logarithms, or longhand multiplication and division. The *change* in g , on the other hand, is readily computed to within 0.1 cm s^{-2} on a small slide rule.)

11-25 A simple pendulum with a supporting steel wire of cross-sectional area 0.01 cm^2 is observed to have a period of 2 s when a 10-kg lead bob is used. The lead bob is replaced by an aluminum bob of the same dimensions having a mass of 2 kg, and the period is remeasured. (a) What was the length of the pendulum with the lead bob? (b) By what fraction is the period changed when the aluminum bob is used? Is it an increase or decrease? [Hint. Use differentials.]

11-26 The balance wheel of a watch vibrates with an angular amplitude of π radians and with a period of 0.5 s. (a) Find its maximum angular velocity. (b) Find its angular velocity when its displacement is one-half its amplitude. (c) Find its angular acceleration when its displacement is 45° .

11-27 A monkey wrench is pivoted at one end and allowed to swing as a physical pendulum. The period is 0.9 s and the pivot is 6 in. from the center of gravity. (a) What is the radius of gyration of the wrench about an axis through the pivot? (b) If the

wrench was initially displaced 0.1 rad from its equilibrium position, what is the angular velocity of the wrench as it passes through the equilibrium position?

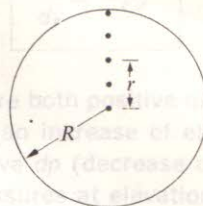


Fig. 11-20

11-28 It is shown in textbooks on mechanics that the moment of inertia I of a body about any axis through any point is given by

$$I = I_G + mh^2,$$

where I_G is the moment of inertia about a *parallel* axis through the center of gravity, m is the mass, and h is the perpendicular distance between the two parallel axes. A solid disk of radius $R = 12 \text{ cm}$ oscillates as a physical pendulum about an axis perpendicular to the plane of the disk at a distance r from its center. (See Fig. 11-20.) (a) Calculate the period of oscillation (for small amplitudes) for the following values of r : 0, $R/4$, $R/2$, $3R/4$, R . (b) Let T_0 represent the period when $r = R$, and T the period at any other value of r . Construct a graph of the dimensionless ratio T/T_0 as a function of the dimensionless ratio r/R . (Note that the graph then describes the behavior of *any* solid disk, whatever its radius.) (c) Prove by the methods of calculus that the period is a minimum when $r = R/\sqrt{2}$. Does this result agree with your graph?

11-29 It is desired to construct a pendulum of period 10 s. (a) What is the length of a *simple* pendulum having this period? (b) Suppose the pendulum must be mounted in a case not over 2 ft high. Can you devise a pendulum, having a period of 10 s, that will satisfy this requirement?

11-30 A meter stick hangs from a horizontal axis at one end and oscillates as a physical pendulum. A body of small dimensions, and of mass equal to that of the meter stick, can be clamped to the stick at a distance h below the axis. Let T represent the period of the system with the body attached, and T_0 the period of the meter stick alone. Find the ratio T/T_0 , (a) when $h = 50 \text{ cm}$, (b) when $h = 100 \text{ cm}$. (c) Is there any value of h for which $T = T_0$? If so, find it and explain why the period is unchanged when h has this value.

11-31 A meter stick is pivoted at one end. At what distance below the pivot should it be struck in order that it start swinging smoothly about the pivot?